SAFA COAST REFINERY

MAINTENANCE & OPERATIONS DEPARTMENT

RELIABILITY INCIDENT REPORT / ARCHIVE COPY

RIR-SC-1990-017 ISSUE DATE: 18 SEPTEMBER 1990

01 EQUIPMENT IDENTIFICATION

Site: Safa Coast | Unit: Crude distillation | System: Feed train
Equipment tag: SC-E-101 | Category: Equipment | Class: Heat exchanger
Asset description: Process heat exchanger | Service: Hydrocarbon
Commissioned: 1982 | Boundary: Equipment unit + dedicated auxiliaries
Subunit: Tube bundle | Failed item: Carbon-steel process tube

02 FAILURE EVENT & OPERATING CONTEXT

Detected: 14-Sep-1990 06:00 | Restored: 16-Sep-1990 06:00 (local)
Operating state before trip: Running, normal load | Event: Forced
Detection: Operator round; visible leak confirmed by pressure test
Failure mode: Tube wall loss | Mechanism: Corrosion
Functional failure: YES | Failure lifecycle stage: I (realized impact)
Impact attribution: F1 - primary actual production-loss event

03 DOWNTIME, PRODUCTION & FINANCIAL IMPACT

Channel	Downtime (days)	Production loss (tonnes)	USD/t
P1	2.00	13,000	44
P2	1.00	4,300	59
P3	0.50	1,050	68

Virtual downtime: MAX(channel days) = 2.00 asset-days
Actual contribution loss: USD 897,100 (1990 money basis)
Maintenance cost: USD 18,500 | Active repair effort: 31.2 hours
Potential loss: NOT ASSESSED | No forecast probability recorded

04 FAILURE INVESTIGATION & RESTORATION

Tube leak caused unit turndown. Inspection found local wall thinning.
Cause: Suspected under-deposit corrosion; laboratory review pending.
Maintenance activity: Corrective. Isolated, opened and plugged tube.
Leak test passed; unit restored at 06:00 on 16 September 1990.
Action: Map remaining tube-wall thickness at next planned shutdown.
Action owner: R. Ahmed | Follow-up due: 30-Sep-1990
No injuries or off-site release reported. Environmental loss: not assessed.

05 RELIABILITY EXPOSURE & DOCUMENT CONTROL

Operating exposure: NOT AVAILABLE - do not calculate MTBF from dates.
Operating temperature: 180 deg C | Pressure: 12 bar(g)
Prepared by: R. Ahmed | Reviewed by: Operations / Inspection
Source: Shift log SC-09/1990, pages 14-16; workshop job SC-0719
Report status in archive: Reviewed | Historical date basis: local time
Laboratory confirmation and cost-invoice references: not available

Check deposit sample before confirming root cause.

R. Ahmed / 18-9-90

ARCHIVE DEMONSTRATION - SYNTHETIC RECORD

Created for iAPM ingestion demonstration in 2026; not a plant record.

MEDFI labels are modern demo annotations for historical migration.